PROBLEM 45

Vibrating Rotor (Unbalanced Turbojet)

Original statement · Problems.pdf, p. 31

Forced Oscillations

To understand the vibrations that occur in a turbojet when the mass distribution of the rotor is slightly unbalanced, we can consider the turbojet as a rigid disk of mass m that rotates at a fixed angular velocity ω, with the center of mass G being separated from the geometric center C by a fixed distance [called ℓ in the figure]. An elastic restoring force (with constant k) tries to keep C at its design position O. We call x_0^G, y_0^G the Cartesian coordinates of G, and x_0^C, y_0^C the Cartesian coordinates of C. Considering the weight of the disk negligible, write down the equation of the dynamics of G. Assuming that the homogeneous solution is damped over time and only the particular solution remains, write down the expression of $x_0^G(t)$ and $x_0^C(t)$. Can we affirm that $|\boldsymbol{OC}| > |\boldsymbol{OG}|$ if $\omega > \sqrt{k/m}$?

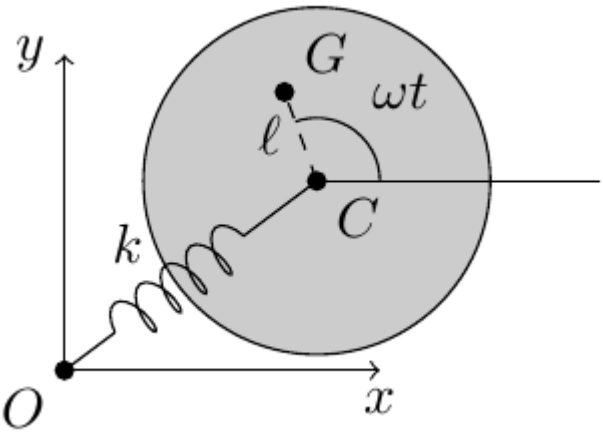


Figure: Rotating disk of centre C with the centre of mass G at distance l, angle omega t from the x direction, spring k from O to C, axes Oxy (Problems.pdf, p. 31).

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a rigid disk of mass m rotates at the constant angular velocity ω. Its geometric centre C is held near the design position O by an elastic restoring force with constant k; its centre of mass G is at the fixed distance ℓ from C (unbalance).
- **Hypotheses:** planar motion in Oxy; the weight is negligible; the spring has zero natural length and acts at C, so its force is $-k\,\mathbf{r}_C$. As the statement asks, the dynamics is that of the centre of mass G, treated as a particle of mass m under the resultant external force (the particle model of Topic 2). The drive that keeps ω constant applies a pure couple, with zero resultant force, so it does not enter the equation of G. The statement also assumes that the homogeneous solution of the equation of G decays, which implies some damping that the model leaves out.
- **Constraints and degrees of freedom:** the disk rotates at the prescribed rate, so its orientation is ωt (a time-dependent, rheonomic constraint) and the segment CG has the fixed length ℓ and the angle ωt. The only free coordinates are the two coordinates of C (or of G): $\mathrm{CDOF} = 2$.
- **Unknowns:** $x_0^G(t)$, $y_0^G(t)$, $x_0^C(t)$, $y_0^C(t)$.
- **Parameters:** m [kg], k [N/m], ℓ [m], ω [rad/s]; natural frequency $\omega_0 = \sqrt{k/m}$.

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with $\mathbf{r}_C = x_0^C\,\mathbf{i}_0 + y_0^C\,\mathbf{j}_0$ and $\mathbf{r}_G = x_0^G\,\mathbf{i}_0 + y_0^G\,\mathbf{j}_0$.
- **Disk-fixed basis.** It is the inertial basis rotated by ωt about $\mathbf{k}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = \begin{pmatrix} \cos^2\omega t + \sin^2\omega t & 0 & 0 \\ 0 & \sin^2\omega t + \cos^2\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

- **Unbalance vector.** The segment CG is fixed in the disk, at the angle ωt from $\mathbf{i}_0$ (first column of the matrix):

$$\mathbf{r}_G = \mathbf{r}_C + \ell\,\mathbf{e}(t), \qquad \mathbf{e}(t) = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$$

Derivation

3.1 Kinematics

Pedagogical Justification: the dynamics refers to G, but the force depends on C, so we express $\mathbf{r}_C$ with the geometry of the rotating segment. The acceleration of G in S_0 is $\ddot{\mathbf{r}}_G = \ddot{x}_0^G\,\mathbf{i}_0 + \ddot{y}_0^G\,\mathbf{j}_0$, and

$$\mathbf{r}_C = \mathbf{r}_G - \ell\,\mathbf{e}(t)$$

3.2 Forces

- **Applied force:** the spring force $-k\,\mathbf{r}_C$, which depends on the position of C, not on that of G. **Constraint forces:** none that affect the centre of mass (the weight is neglected and the drive is a pure couple). **Inertia forces:** none, S_0 is inertial.

3.3 Equations of Motion

Pedagogical Justification: Newton's second law for G with $\mathbf{r}_C = \mathbf{r}_G - \ell\,\mathbf{e}(t)$ gives a forced linear equation (Notes Sec. 5.4, oscillator under a periodic force):

$$m\ddot{\mathbf{r}}_G = -k\,\mathbf{r}_C = -k\,[\mathbf{r}_G - \ell\,\mathbf{e}(t)]$$

- Dividing by m with $\omega_0^2 = k/m$, and eliminating $\mathbf{r}_C$ through the geometric relation of 3.1 (so the unknowns reduce to the two coordinates of G), the two components decouple (two unknowns, two equations):

$$\ddot{x}_0^G + \omega_0^2 x_0^G = \omega_0^2 \ell\cos\omega t, \qquad \ddot{y}_0^G + \omega_0^2 y_0^G = \omega_0^2 \ell\sin\omega t$$

3.4 Integration and Answers

Particular solution for G

> **Pedagogical Justification:** the general solution is the homogeneous one, $A\cos\omega_0 t + B\sin\omega_0 t$, which the statement assumes damped out, plus a particular solution. Since the forcing is harmonic at ω, we try the same harmonic with an unknown amplitude (undetermined coefficients): $x_0^G = a\cos\omega t$, so $\ddot{x}_0^G = -\omega^2 a\cos\omega t$.

$$-\omega^2 a + \omega_0^2 a = \omega_0^2 \ell \implies a = \frac{\omega_0^2 \ell}{\omega_0^2 - \omega^2} \qquad (\omega \neq \omega_0)$$

- The same steps with $y_0^G = b\sin\omega t$, $\ddot{y}_0^G = -\omega^2 b\sin\omega t$, give $-\omega^2 b + \omega_0^2 b = \omega_0^2 \ell$, so $b = a$:

$$\boxed{x_0^G(t) = \frac{\omega_0^2 \ell}{\omega_0^2 - \omega^2}\cos\omega t}, \qquad y_0^G(t) = \frac{\omega_0^2 \ell}{\omega_0^2 - \omega^2}\sin\omega t$$

Position of the geometric centre

- From $\mathbf{r}_C = \mathbf{r}_G - \ell\,\mathbf{e}(t)$, component by component:

$$\boxed{x_0^C(t) = \left[\frac{\omega_0^2}{\omega_0^2 - \omega^2} - 1\right]\ell\cos\omega t = \frac{\omega^2 \ell}{\omega_0^2 - \omega^2}\cos\omega t}, \qquad y_0^C(t) = \frac{\omega^2 \ell}{\omega_0^2 - \omega^2}\sin\omega t$$

Is $|OC| > |OG|$ for $\omega > \sqrt{k/m}$?

- Both points describe circles about O at the rotation rate ω, with radii

$$|OG| = \frac{\omega_0^2 \ell}{|\omega_0^2 - \omega^2|}, \qquad |OC| = \frac{\omega^2 \ell}{|\omega_0^2 - \omega^2|}, \qquad \frac{|OC|}{|OG|} = \frac{\omega^2}{\omega_0^2} = \frac{m\omega^2}{k}$$

- **Yes.** The ratio exceeds 1 exactly when $\omega > \omega_0 = \sqrt{k/m}$ (the case $\omega = \omega_0$ is excluded: the bounded harmonic particular solution $a\cos\omega t$ does not exist there, and the response grows linearly in time, $x_0^G = \tfrac{1}{2}\omega_0 \ell\, t\sin\omega_0 t$, which is the resonance: with $x_p = Dt\sin\omega_0 t$,

$\ddot{x}_p = 2D\omega_0 \cos\omega_0 t - D\omega_0^2 t \sin\omega_0 t$, so $\ddot{x}_p + \omega_0^2 x_p = 2D\omega_0 \cos\omega_0 t = \omega_0^2 \ell \cos\omega_0 t$ and $D = \omega_0 \ell / 2$). For $\omega < \omega_0$ the opposite holds.

Interpretation & Checks

- **Slow rotation ($\omega \to 0$):** $x_0^G \to \ell$ and $x_0^C \to 0$: C sits at O and G at distance ℓ, as for a static unbalanced disk.
- **Fast rotation ($\omega \gg \omega_0$):** $|OG| \to \ell\omega_0^2/\omega^2 \to 0$ and $|OC| \to \ell$: above the critical speed the disk self-centres and spins about its centre of mass, with C circling G at radius ℓ and in antiphase with the unbalance (the factor $\omega_0^2 - \omega^2$ is negative).
- **Resonance ($\omega = \omega_0$):** the amplitudes diverge in this undamped model.
- **Dimensional check:** $\left[\omega_0^2 \ell / (\omega_0^2 - \omega^2)\right] = \mathrm{m}$ and $[k/m] = \mathrm{N\,m^{-1}kg^{-1}} = \mathrm{s^{-2}} = [\omega^2]$.
- **Check against the official key:** the key of `Problems.pdf` gives, for $\omega \neq \omega_0$, $x_0^G(t) = \frac{\omega_0^2 \ell}{\omega_0^2 - \omega^2} \cos\omega t$ and $x_0^C(t) = \frac{\omega^2 \ell}{\omega_0^2 - \omega^2} \cos\omega t$, with $\omega_0 = \sqrt{k/m}$, identical to the results above. Sympy confirms that the particular solution satisfies the equations with zero residual.

Source: Problems.pdf